GlueForge

A complete dynamics studio — bus glue, drum & vocal control, mastering, multiband, mid/side, external sidechain, and a tempo-synced EDM pump with a hand-drawn shape.

VST3 & Standalone · VCA · FET · Opto · 3-band Multiband · Lookahead + Oversampling

Contents

NEW HERE?

Click the `?` button in the plugin to toggle hover tooltips on every control, and the `Manual` button to open this document any time. Jump to **§1 Quick Start** to make a sound in 60 seconds.

1 Quick Start

Three ways in, depending on what you're reading for.

Glue a bus

1. Load on a drum or mix bus.
2. **Ratio** 2:1, **Knee** soft.
3. Lower **Threshold** for 2–4 dB GR.
4. **Attack** 30 ms, **Release** auto-feel (~150 ms).
5. **Character** = **VCA**. Done.

EDM pump

1. Set **Trigger** = **Tempo Duck**.
2. Pick a **Duck Rate** (1/4).
3. Dial **Depth** to taste.
4. Shape the curve in the center display.
5. No sidechain track needed.

Classic sidechain

1. Set **Trigger** = **External SC**.
2. Route the kick to GlueForge's sidechain input.
3. Fast **Attack**, medium **Release**.
4. Use **SC HPF** to ignore sub rumble.
5. **Listen** to audition the trigger.

2 Installing & Loading

Install

Run `GlueForge-Setup-x.y.z.exe`. It places the VST3 in `C:\Program Files\Common Files\VST3` (where every DAW scans) and, optionally, the standalone app.

Load in Ableton Live

1. Open **Preferences** → **Plug-Ins** and make sure **Use VST3 System Folders** is on, then **Rescan**.
2. Find **GlueForge** under *Plug-Ins* → *Nightshift Audio* and drop it on a track.

AFTER AN UPDATE
A running DAW keeps the old plug-in cached in memory. Fully **quit and relaunch** your DAW, then delete and re-add the instance to load the new version.

3 Interface Tour

Two tabs share one window: **Compressor** (the main engine) and **Multiband** (three independent bands).

Top bar

- **Preset menu** + `<` `>` to step through factory presets.
- **Save / Load** your own presets as `.xml` files.
- **A / B** hold two full settings; **A>B** copies one into the other.
- **Bypass** the whole plug-in.

Tabs row

- **COMPRESSOR** / **MULTIBAND** switch the body.
- `?` toggles **hover help** on every control.
- `Manual` opens this PDF in your system viewer.

The compressor page at a glance

ZONE	WHAT IT SHOWS
Meters (IN · GR · OUT)	Input level, gain reduction, and output level — peak-reading with a 36 dB scale.
Center display	The **transfer curve** with a live operating dot — or, in Tempo Duck mode, the editable **pump shape**.
GR history	A scrolling graph of gain reduction over time, so you can see the compressor breathe.
Control rows	Five grouped sections: Dynamics · Detector/Output · Sidechain · Tempo Duck/Character · Mix/Quality.

TIP
Knobs respond to **vertical or horizontal drag**. Double-click a knob to type a value, and hold **Ctrl** (Cmd on Mac) for fine adjustment.

4 Dynamics

The core compressor: when it grabs, how hard, how fast, and how it lets go.

CONTROL	RANGE	WHAT IT DOES
Threshold	`−60…0 dB`	Level above which compression begins. Lower it for more gain reduction.
Ratio	`1:1…20:1`	How hard signal above threshold is reduced. 2:1 = gentle glue, 4:1 = control, 10:1+ = limiting.
Knee	`0…24 dB`	Width of the soft transition around the threshold. Higher = smoother, more gradual onset.
Attack	`0.05…200 ms`	How fast it clamps once over threshold. Fast catches transients; slow lets punch through.
Release	`5…2000 ms`	How fast it recovers after the signal drops. Short = pumpy/energetic, long = smooth/transparent.
Hold	`0…500 ms`	Keeps full reduction for a set time before release begins — steadies fast, choppy material.

Attack & release, intuitively

Short attack tames spikes but dulls punch. Long attack preserves the initial hit and compresses the body. Short release adds energy and "pump"; long release is invisible and gluey. When in doubt, start at **10 ms / 120 ms** and adjust by ear while watching the GR meter.

Reading the transfer curve

The diagonal line is your in→out mapping. Below threshold it's 1:1 (45°); above it bends by the ratio, rounded by the knee. The moving dot is the **live operating point** — where your signal sits on the curve right now.

5 Detector & Output

How the compressor measures level, and how you make up the loss.

CONTROL	RANGE	WHAT IT DOES
Detect	`Peak…RMS`	Blends between fast **Peak** (transient-accurate) and **RMS** (loudness-like, smoother). Default 20% RMS.
Max GR	`0…60 dB / Off`	Caps the maximum gain reduction so the compressor can't clamp past a set amount. **Off** = unlimited.
Link	`0…100%`	Stereo linking. 100% = both channels share one detector (stable image); 0% = fully independent (dual-mono).
Makeup	`−12…+24 dB`	Manual gain to compensate for the level lost to compression.
Output	`−24…+24 dB`	Final output trim, after everything.
Auto	`on/off`	Auto-makeup: estimates and applies makeup gain automatically as you change threshold/ratio.

GAIN STAGING
Use **Auto** for quick level-matched A/B while dialing settings, then switch it off and set **Makeup/Output** by ear for the final balance. Level-matched comparisons keep "louder" from fooling you into "better".

Peak vs RMS, in practice

Toward Peak

Catches every transient. Great for limiting, drum control, and anything where the spikes are the problem.

Toward RMS

Responds to average loudness like the ear does. Smoother, more musical glue on busses, vocals, and masters.

6 Sidechain

The **Trigger** selector decides what drives the compressor.

TRIGGER MODE	WHAT DRIVES COMPRESSION
Internal	The track's own signal (a normal compressor).
External SC	An external source routed to the plug-in's sidechain input — classic ducking (e.g. kick ducks bass).
Tempo Duck	A host-synced internal pump — no external track needed (see §7).

Detector filters

CONTROL	RANGE	WHAT IT DOES
SC HPF	`20…2000 Hz`	High-pass on the *detection* signal. Raise it so deep bass doesn't trigger reduction (the audio stays full-range).
SC LPF	`200 Hz…20 kHz`	Low-pass on the detection signal. Lower it to make the compressor ignore highs/sibilance.
Listen	`on/off`	Audition the (filtered) detection signal so you can hear exactly what the compressor reacts to.

ROUTING IN ABLETON
On the track with GlueForge, the sidechain source is selected in your DAW's plug-in sidechain dropdown. In Ableton, expand the device's title bar **Sidechain** section and pick the kick (or other source) as the audio-from input, then set **Trigger = External SC**.

De-essing trick

Set **Trigger = External SC** (or Internal), push **SC HPF** up around 4–6 kHz, fast **Attack**, and the compressor will react mainly to sibilance — a quick, surgical de-esser.

7 Tempo Duck & the Pump-Shape Editor

Set **Trigger = Tempo Duck** and the center display becomes a drawable pump shape, locked to your host tempo.

CONTROL	RANGE	WHAT IT DOES
Duck Rate	`note divisions`	The cycle length of one pump, synced to the project tempo (e.g. 1/4, 1/8, dotted, triplet).
Depth	`0…36 dB`	How deep each pump dips at its lowest point.

Editing the shape

Move

Drag any node to reposition it in time (x) and level (y).

Bend

Drag the *body of a segment* (between two nodes) to curve it — from linear to sharp exponential.

Add / Remove

Double-click empty space to add a node; double-click a node to delete it.

The shape is one tempo cycle, looping. The left edge is the start of the beat (deepest duck for a classic sidechain feel), recovering back to full level by the right edge. Curved segments give you the snappy, exponential "breathe" that linear ramps can't.

CARRIED EVERYWHERE
The pump shape travels with your project, with each **preset**, and with **A/B** — so a custom curve is never lost.

Why Tempo Duck instead of a real sidechain?

It's tighter and repeatable: no need to route a ghost kick, no level drift, and the exact shape is yours to draw. Perfect for EDM/pop pumping on pads, bass, and whole busses.

8 Character & Saturation

Three classic compressor "flavours", plus a saturation stage for colour and harmonics.

VCA

Clean, fast, punchy and transparent. The modern bus-glue sound — controlled and tight without obvious colouration.

FET

Aggressive and quick with a forward, harmonically rich edge. Excellent on drums, room mics, and vocals that need attitude.

Opto

Smooth and program-dependent — the gentle, rounded behaviour of a light-based compressor. Beautiful on vocals, bass, and master glue.

Saturation

CONTROL	RANGE	WHAT IT DOES
Drive	`0…100%`	How hard the signal is pushed into the character stage — more harmonics, more density.
Sat Mix	`0…100%`	Blends the saturated signal back in. 0% = clean dynamics only; raise for warmth and weight.

SUBTLE IS PLENTY
A little goes a long way. Try **Drive 30%** with **Sat Mix 15–25%** for analog-style glue that you feel more than hear.

9 Mix & Quality

Parallel blend, mid/side processing, and the precision controls.

CONTROL	RANGE	WHAT IT DOES
Mix	`0…100%`	Dry/wet blend for **parallel compression**. Keep transients (dry) while adding compressed body (wet).
Look	`0…10 ms`	Lookahead — lets the compressor "see" transients early for cleaner catching. Reported to the host as latency (PDC).
OS	`Off · 2× · 4× · 8×`	Oversampling reduces aliasing from fast compression and saturation. Higher = cleaner, more CPU. Adds latency (PDC).
M/S	`on/off`	Process **Mid/Side** instead of Left/Right — compress the center and sides independently of each other.

Parallel (New York) compression

Crush hard — low threshold, high ratio, fast attack — then pull **Mix** back to ~30–50%. You get the power of heavy compression with the life of the untouched signal.

Mid/Side on a master

Enable **M/S** to glue the center (vocals, kick, bass) without squashing the stereo sides — or vice-versa. Pairs well with the Multiband tab.

LATENCY
Lookahead and oversampling add latency, which GlueForge reports to your DAW for automatic delay compensation. For tracking/live monitoring, keep both at their lowest.

10 Multiband Tab

Split the signal into **Low · Mid · High** and compress each independently — with an EQ-style display you can drag.

The display

- Log-frequency graph from 20 Hz to 20 kHz with three shaded band regions.
- **Drag the crossover lines** to set where bands split.
- Each band shows a **live gain-reduction shelf** — it visibly dips by how hard that band is compressing.

CONTROL	RANGE	WHAT IT DOES
Multiband ON	`on/off`	Switches the engine from single-band to 3-band multiband.
X Low	`30…1000 Hz`	Low|Mid crossover frequency (also draggable on the graph).
X High	`1…16 kHz`	Mid|High crossover frequency (also draggable on the graph).
Solo	`None/Low/Mid/High`	Hear one band on its own to dial it in.
Thr / Ratio (per band)	`−60…0 dB / 1…20:1`	Independent threshold and ratio for each band.
Trim (per band)	`−24…+24 dB`	Per-band output level after compression — a built-in tonal balance.
Bypass (per band)	`on/off`	Skip compression on a band while keeping it in the mix.

CROSSOVERS RECONSTRUCT FLAT
GlueForge uses Linkwitz-Riley crossovers with allpass compensation, so with no compression the three bands sum back to the original signal — no phasey comb-filtering at the splits.

Common multiband moves

Tame a boomy low end (compress **Low** only), de-harsh a bright mix (gentle **High**), or control a vocal's body without dulling air (compress **Mid**, leave **High** open).

11 Metering & Displays

IN / OUT

Peak input and output levels on a 36 dB scale. Watch OUT to keep your gain staging in check.

GR meter

How much the compressor is pulling down right now, in dB. Aim for musical amounts — often 2–6 dB for glue.

GR history

A scrolling timeline of gain reduction so you can see the rhythm of the compression, not just the peak.

12 Presets & A/B

Presets

Factory presets are grouped by category in the menu; step through them with `<` and `>`. **Save** writes your current settings (including the pump shape) to an `.xml` file under `Documents\Nightshift Audio\GlueForge\Presets`; **Load** recalls one.

A/B compare

A and **B** are two independent snapshots of every parameter. Tweak under A, click **B** to branch a variation, and flip between them to compare. **A>B** copies the current slot into the other so you can start a variation from where you are.

EVERYTHING IS STATE
Presets, A/B, and your DAW's project all capture the full plug-in state — knobs, tabs, multiband, and the hand-drawn pump shape alike.

13 Recipes

Starting points, not rules. Trust your ears and the GR meter.

Drum bus glue

VCA · Ratio 2–4:1 · Attack 20–30 ms (let the snap through) · Release ~150 ms · 3–5 dB GR · a touch of Drive.

Lead vocal

Opto · Ratio 3:1 · soft Knee · Attack 5–10 ms · Release 80–150 ms · Detect toward RMS · Max GR ~6 dB to keep it natural.

Parallel drum smash

FET · low Threshold · Ratio 10:1 · fast Attack · fast Release · then **Mix ~35%**.

Mix-bus master glue

VCA or Opto · Ratio 2:1 · Knee high · Attack 30 ms · Release auto-feel · 1–2 dB GR · consider **M/S** + light Multiband.

EDM pump (no sidechain)

Trigger Tempo Duck · Rate 1/4 · Depth to taste · draw a fast-down / curved-up shape · apply to pads, bass, and busses.

Boom control

Multiband ON · compress **Low** only · raise X Low to ~150 Hz · gentle ratio · leave Mid/High bypassed.

GOLDEN RULE
Set it, then **bypass** and compare at matched loudness. If the bypassed version sounds better, back off. Compression should serve the song, not the meter.

14 Parameter Reference

Every parameter, its range, and default.

Compressor

PARAMETER	RANGE	DEFAULT
Threshold	−60…0 dB	−18 dB
Ratio	1:1…20:1	4:1
Knee	0…24 dB	6 dB
Attack	0.05…200 ms	10 ms
Release	5…2000 ms	120 ms
Hold	0…500 ms	0 ms
Detect	Peak…RMS	20% RMS
Max GR	0…60 dB / Off	Off
Link	0…100%	100%
Makeup	−12…+24 dB	0 dB
Auto Makeup	on/off	off
Output	−24…+24 dB	0 dB

Sidechain & Duck

PARAMETER	RANGE / DEFAULT
Trigger	Int / Ext SC / Duck · Int
SC HPF	20…2000 Hz · 20
SC LPF	200…20k Hz · 20k
Listen	on/off · off
Duck Rate	divisions · 1/4
Depth	0…36 dB · 12

Character & Quality

PARAMETER	RANGE / DEFAULT
Character	VCA/FET/Opto · VCA
Drive	0…100% · 30%
Sat Mix	0…100% · 0%
Mix	0…100% · 100%
Lookahead	0…10 ms · 0
Oversampling	Off/2×/4×/8× · Off
Mid/Side	on/off · off

Multiband

PARAMETER	RANGE	DEFAULT
Multiband	on/off	off
X Low / X High	30…1000 Hz / 1…16 kHz	200 Hz / 2.5 kHz
Per-band Thr	−60…0 dB	−18 dB
Per-band Ratio	1…20:1	4:1
Per-band Trim	−24…+24 dB	0 dB
Solo	None/Low/Mid/High	None